

Assignment 4 a

Title: Implementation of jQuery Mobile

Problem Statement: Create a simple Mobile Website using jQuery Mobile.

Objective: Apply jQuery Mobile library conventions and demonstrate a website.

S/W Packages and H/W apparatus used:

Software/Language/Libraries used: HTML, jQuery Mobile Linux

OS: Ubuntu/Windows, VSCode editor.

PC with the configuration as Pentium IV 1.7 GHz. 128M.B RAM, 40 G.B HDD, 15’’Color Monitor, Keyboard, Mouse.

References:

1. <https://jquerymobile.com/>
2. <https://demos.jquerymobile.com/1.4.5/>

Theory:

What is Mobile jQuery?

jQuery Mobile is the easiest way to build sites and apps that are accessible on all popular smartphone, tablet and desktop devices.

jQuery Mobile is a touch-optimized HTML5 UI framework designed to make responsive web sites and apps that are accessible on all smartphone, tablet and desktop devices.

jQuery Mobile has multiple Pages and Navigation, Widgets, Form Widgets etc.

1. Pages

The page is the primary unit of interaction in jQuery Mobile and is used to group content into logical views that can be animated in and out of view with page transitions. A HTML document may start with a single "page" and the Ajax navigation system will load additional pages on demand into the DOM as users navigate around. Alternatively, a HTML document can be built with multiple "pages" inside it and the framework will transition between these local views with no need to request content from the server.

Mobile page structure

A jQuery Mobile site must start with an HTML5 doctype to take full advantage of all of the framework's features. (Older devices with browsers that don't understand HTML5 will safely ignore the 'doctype' and various custom attributes.)

In the head, references to jQuery, jQuery Mobile and the mobile theme CSS are all required to start things off. The easiest way to get started is to link to files hosted on the jQuery CDN or for best performance, build a custom bundle.

```
<!DOCTYPE html>
<html>
<head>
  <title>Page Title</title>
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://code.jquery.com/mobile/[version]/jquery.mobile-[version].min.css" />
  <script src="https://code.jquery.com/jquery-[version].min.js"></script>
  <script src="https://code.jquery.com/mobile/[version]/jquery.mobile-
[version].min.js"></script>
</head>

<body>
  ...content goes here...
```

```
</body>
</html>
```

Viewport meta tag

Note above that there is a meta viewport tag in the head to specify how the browser should display the page zoom level and dimensions. If this isn't set, many mobile browsers will use a "virtual" page width around 900 pixels to make it work well with existing desktop sites but the screens may look zoomed out and too wide. By setting the viewport attributes to `content="width=device-width, initial-scale=1"`, the width will be set to the pixel width of the device screen.

```
<meta name="viewport" content="width=device-width, initial-scale=1">
```

These settings do not disable the user's ability to zoom the pages, which is nice from an accessibility perspective. There is a minor issue in iOS that doesn't properly set the width when changing orientations with these viewport settings, but this will hopefully be fixed in a future release. You can set other viewport values to disable zooming if required since this is part of your page content, not the library.

Inside the body: Pages

Inside the `<body>` tag, each view or "page" on the mobile device is identified with an element (usually a `div`) with the `data-role="page"` attribute.

```
<div data-role="page">
    ...
</div>
```

Within the "page" container, any valid HTML markup can be used, but for typical pages in jQuery Mobile, the immediate children of a "page" are `div`s with `data-role="header"`, `class="ui-content"`, and `data-role="footer"`.

```
<div data-role="page">
    <div data-role="header">...</div>
```

```
<div role="main" class="ui-content">...</div>
<div data-role="footer">...</div>
</div>
```

Putting it together: Basic single page template

Putting it all together, this is the standard boilerplate page template you should start with on a project:

```
<!DOCTYPE html>
<html>
<head>
  <title>Page Title</title>

  <meta name="viewport" content="width=device-width, initial-scale=1">

  <link rel="stylesheet"
href="https://code.jquery.com/mobile/[version]/jquery.mobile-[version].min.css" />
  <script src="https://code.jquery.com/jquery-[version].min.js"></script>

  <script src="https://code.jquery.com/mobile/[version]/jquery.mobile-
[version].min.js"></script>
</head>
<body>

<div data-role="page">

  <div data-role="header">
    <h1>Page Title</h1>
  </div><!-- /header -->

  <div role="main" class="ui-content">
    <p>Page content goes here.</p>
```

```
</div><!-- /content -->

<div data-role="footer">
    <h4>Page Footer</h4>
</div><!-- /footer -->
</div><!-- /page -->

</body>
</html>
```

Multi-page template structure

A single HTML document can contain multiple "pages" that are loaded together by stacking multiple divs with a data-role of "page". Each "page" block needs a unique id (id="foo") that will be used to link internally between "pages" (href="#foo"). When a link is clicked, the framework will look for an internal "page" with the id and transition it into view.

Here is an example of a two "page" site built with two jQuery Mobile divs navigated by linking to an id placed on each page wrapper. Note that the ids on the page wrappers are only needed to support the internal page linking, and are optional if each page is a separate HTML document. Here is what two pages look like inside the body element.

```
<body>

<!-- Start of first page -->
<div data-role="page" id="foo">

    <div data-role="header">
        <h1>Foo</h1>
    </div><!-- /header -->

    <div role="main" class="ui-content">
```

```
<p>I'm first in the source order so I'm shown as the page.</p>
<p>View internal page called <a href="#bar">bar</a></p>
</div><!-- /content -->

<div data-role="footer">
  <h4>Page Footer</h4>
</div><!-- /footer -->
</div><!-- /page -->

<!-- Start of second page -->
<div data-role="page" id="bar">

  <div data-role="header">
    <h1>Bar</h1>
  </div><!-- /header -->

  <div role="main" class="ui-content">
    <p>I'm the second in the source order so I'm hidden when the page loads.
    I'm just shown if a link that references my id is beeing clicked.</p>
    <p><a href="#foo">Back to foo</a></p>
  </div><!-- /content -->

  <div data-role="footer">
    <h4>Page Footer</h4>
  </div><!-- /footer -->
</div><!-- /page -->
</body>
```

Ajax Navigation

The \$.mobile.navigate method and the navigate event form the foundation of jQuery Mobile's navigation infrastructure. As such, they can function outside the confines of jQuery Mobile as a clean and intuitive navigation/history API. Introduction

jQuery Mobile includes a navigation system to load pages into the DOM via Ajax, enhance the new content, then display [pages](#) with a rich set of animated [transitions](#). The navigation system uses progressive enhancement to automatically 'hijack' standard links and form submissions and route them as an Ajax request.

One of jQuery Mobile's core features is the ability to load and view content from disparate pages into the initial document with support for standard navigation methods like anchors and the back button. To accomplish this the library has progressive support for hashchange and popstate coupled with internal history tracking which can be used à la carte.

An example use case would be something like Twitter's web client. The first step is to hijack link clicks on the page and use the URL that represents that UI state to track history with \$.mobile.navigate. It's at this point that any additional information about the UI necessary for operation on return using the back button would be stored (see, foo property of the object argument to the navigate method).

```
// Define a click binding for all anchors in the page
$( "a" ).on( "click", function( event ) {

    // Prevent the usual navigation behavior
    event.preventDefault();

    // Alter the url according to the anchor's href attribute, and
    // store the data-foo attribute information with the url
    $.mobile.navigate( $(this).attr( "href" ), {
        foo: $(this).attr("data-foo")
    });
});
```

```
// Hypothetical content alteration based on the url. E.g, make //  
an Ajax request for JSON data and render a template into the page.  
    alterContent( $(this).attr("href") );  
  
});
```

Next, a `navigate` event binding helps in responding to backward and forward navigation via the browsers history API. Here the `alterContent` function can address the direction in which the browser is navigating as well as any additional information stored on the `data` object when `$.mobile.navigate` was invoked to store the corresponding history entry.

```
// Respond to back/forward navigation  
$( window ).on( "navigate", function( event, data ){  
    if ( data.state.foo ) {  
  
        // Make use of the arbitrary data stored  
  
    }  
  
    if ( data.state.direction == "back" ) {  
  
        // Make use of the directional information  
  
    }  
  
    // reset the content based on the url  
    alterContent( data.state.url );  
  
});
```

2. Widgets

Navbar

jQuery Mobile has a very basic navbar widget that is useful for providing up to 5 buttons with optional icons in a bar.

Navbar basics

A navbar is coded as an unordered list of links wrapped in a container element that has the data-role="navbar" attribute. When a link in the navbar is clicked it gets the active (selected) state. The ui-btn-active class is first removed from all anchors in the navbar before it is added to the activated link. If this is a link to another page, the class will be removed again after the transition has completed.

To set an item to the active state, add class="ui-btn-active" to an anchor in the markup. Additionally add a class of ui-state-persist to make the framework restore the active state each time the page is shown while it exists in the DOM.

Navbars with 1 item will render as 100% wide.

```
<div data-role="navbar">
  <ul>
    <li><a href="#" class="ui-btn-active">One</a></li>
  </ul>
</div>
```

The navbar items are set to divide the space evenly so in this case, each button is 1/2 the width of the browser window:

```
<div data-role="navbar">
  <ul>
    <li><a href="#" class="ui-btn-active">One</a></li>
    <li><a href="#">Two</a></li>
  </ul>
</div>
```

And there are also ways to add third and fourth items.

3. Form Widgets

Checkbox

Checkbox inputs are used to provide a list of options where more than one can be selected. Checkbox buttons are enhanced by the checkboxradio widget.

Basic markup

To create a single checkbox, add an input with a type="checkbox" attribute and a corresponding label. If the input isn't wrapped in its corresponding label, be sure to set the for attribute of the label to match the id of the input so they are semantically associated.

```
<form>
  <label>
    <input type="checkbox" name="checkbox-0">Check Me
  </label>
</form>
```

Radio buttons

Radio inputs are used to provide a list of options where only a single option can be selected. Radio buttons are enhanced by the checkboxradio widget.

Basic markup

To create a set of radio buttons, add an input with a type="radio" attribute and a corresponding label. Set the for attribute of the label to match the id of the input so they are semantically associated.

```
<form>
  <label>
    <input type="radio" name="radio-choice-0">One
  </label>
  <label for="radio-choice-0b">Two</label>
    <input type="radio" name="radio-choice-0" id="radio-choice-0b" class="custom">
</form>
```

Sample Code:**Index.html**

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta http-equiv="X-UA-Compatible" content="IE=edge">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">


  <link rel="stylesheet" href="https://code.jquery.com/mobile/1.4.5/jquery.mobile-1.4.5.min.css" />

  <script src="https://code.jquery.com/jquery-1.11.1.min.js"></script>
```

```
<script src="https://code.jquery.com/mobile/1.4.5/jquery.mobile-1.4.5.min.js"></script>

<title>Demo of jQuery Mobile</title>

</head>

<body>

  <div data-role="page" style="background-color:aliceblue;">

    <div data-role="header" data-theme="b">

      <h1>My Website</h1>

    </div>

    <div data-role="navbar">

      <ul>

        <li><a href="index.html">Home</a></li>

        <li><a href="about.html">About</a></li>

        <li><a href="login_register.html">Register for Newsletter</a></li>

      </ul>

    </div>

    <center><h3>Gallery</h3></center>

    <div class="content">

      <center>

        <div class="rows">

          <img

            width="240" height="200">

          <img

            width="240" height="200">

        </div>

      </center>

    </div>

  </div>

</body>

</html>
```

```
<img alt="Img4" width="240" height="200">

<img width="240" height="200">



</div>

<br>

<div class="rows">

    src="img1.jpg" alt="Img1"
    src="img2.jpg" alt="Img2"
    src="img3.jpg" alt="Img3"
    src="img4.jpg" alt="Img4"


</div>

<br>

<div class="rows">


</div>

<img width="240" height="200">

<img width="240" height="200">

<img width="240" height="200">
```

```
<img width="240" height="200">
```

```
<img width="240" height="200">
```

```
<br>
```

```
<div class="rows">
```

```

```

```

```

```

```

```

```

```

```

```

```

```
</div>
```

```
<br>
```

```
<div class="rows">
```

```
src="img1.jpg" alt="Img1"
```

```
src="img2.jpg" alt="Img2"
```

```
src="img3.jpg" alt="Img3"
```

```
src="img4.jpg" src="img5.jpg"
```

```
alt="Img4"
```

```

```

```
</div>
```

```
<br>
```

```
<img width="240" height="200">
```

```
<img width="240" height="200">
```

```
<img width="240" height="200">
```

```
<img alt="Img4" width="240" height="200">

<img width="240" height="200">
<div class="rows">


</div>

<br>

<div class="rows">


</div>

<br>

<div class="rows">

  <img width="240" height="200">
  <img width="240" height="200">
  <img width="240" height="200">
```

```
<img          width="240" height="200">  
<img          width="240" height="200">  
  
    src="img2.jpg" alt="Img2"  
    src="img3.jpg" alt="Img3"  
    src="img4.jpg" alt="Img4"
```

```
<img          width="240" height="200">  
<img          width="240" height="200">  
<img          width="240" height="200">
```

```
<img alt="Img4" width="240" height="200">
```

```
<img width="240" height="200">  
src="img5.jpg"  
src="img6.jpg" alt="Img4"
```

```
</div>
```

```
<br>
```

```
<div class="rows">
```

```

```

```

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```

```

```

```

```

```

```

```

```
</div>
```

```
<br>
```

```
<div class="rows">
```

```

```

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```

```

```

```

```

```
<img width="240" height="200">
```

```
<img width="240" height="200">
```

```
<img width="240" height="200">
```

```
</div>
```

```
<br>
```

```
<div class="rows">
```

```

```

```

```

```
src="img3.jpg" alt="Img3"
```

```
src="img4.jpg" alt="Img4"
```

```
src="img5.jpg" alt="Img4"
```

```
src="img6.jpg"
```

```
</div>
```

```
<br>
```

```
<div class="rows">
```

```

```

```

```

```

```

```

```

```

```

```

```

```
</div>
```

```
<br>
```

```
<div class="rows">
```

```

```

```

```

```

```

```

```

```
<img alt="Img4" width="240" height="200">
  



</div>

<br>

</center>

</div>

<div data-role="footer">

  <center>

    <a href="index.html" class="ui-btn ui-btn-inline">Home</a>

    <a href="about_us.html" class="ui-btn ui-btn-inline">About</a>

    <a href="login_register.html" class="ui-btn ui-btn-inline">Register</a>
  </center>
</div>

</div>

</body>

</html>
```

Register.html

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta http-equiv="X-UA-Compatible" content="IE=edge">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<link rel="stylesheet" href="https://code.jquery.com/mobile/1.4.5/jquery.mobile-1.4.5.min.css" />

<script src="https://code.jquery.com/jquery-1.11.1.min.js"></script>

<script src="https://code.jquery.com/mobile/1.4.5/jquery.mobile-1.4.5.min.js"></script>

<title>Demo of jQuery Mobile</title>

</head>

<body>

<div data-role="page" style="background-color:aliceblue;">

  <div data-role="header" data-theme="b">

    <h1>My Website</h1>

  </div>

  <div data-role="navbar">

    <ul>

      <li><a href="index.html">Home</a></li>

      <li><a href="about.html">About</a></li>

      <li><a href="login_register.html">Register for Newsletter</a></li>

    </ul>

  </div>

</div>
```

</div>

<div class="content" style="width:50%; margin: 5% auto;">

<center><h3>Register</h3></center>

<div data-role="content">

<label for="userName">Username</label>

<input type="text" name="userName">

<label for="email">Email</label>

<input type="text" name="email">

<label for="password">Password</label>

<input type="text" name="password">

<center>Register

Me</center>

</div>

</div>

</div>

</body>

</html>

About.html

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta http-equiv="X-UA-Compatible" content="IE=edge">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

```
<link rel="stylesheet" href="https://code.jquery.com/mobile/1.4.5/jquery.mobile-1.4.5.min.css" />

<script src="https://code.jquery.com/jquery-1.11.1.min.js"></script>
<script src="https://code.jquery.com/mobile/1.4.5/jquery.mobile-1.4.5.min.js"></script>

<title>Demo of jQuery Mobile</title>

</head>

<body>

<div data-role="page" style="background-color:aliceblue;">

  <div data-role="header" data-theme="b">

    <h1>My Website</h1>

  </div>

  <div data-role="navbar">

    <ul>

      <li><a href="index.html">Home</a></li>

      <li><a href="about.html">About</a></li>

      <li><a href="login_register.html">Register for Newsletter</a></li>

    </ul>

  </div>

  <center><h3>Gallery</h3></center>

  <div class="content">

    <center>

      We have a HD image gallery. All the images are taken from https://unsplash.com/

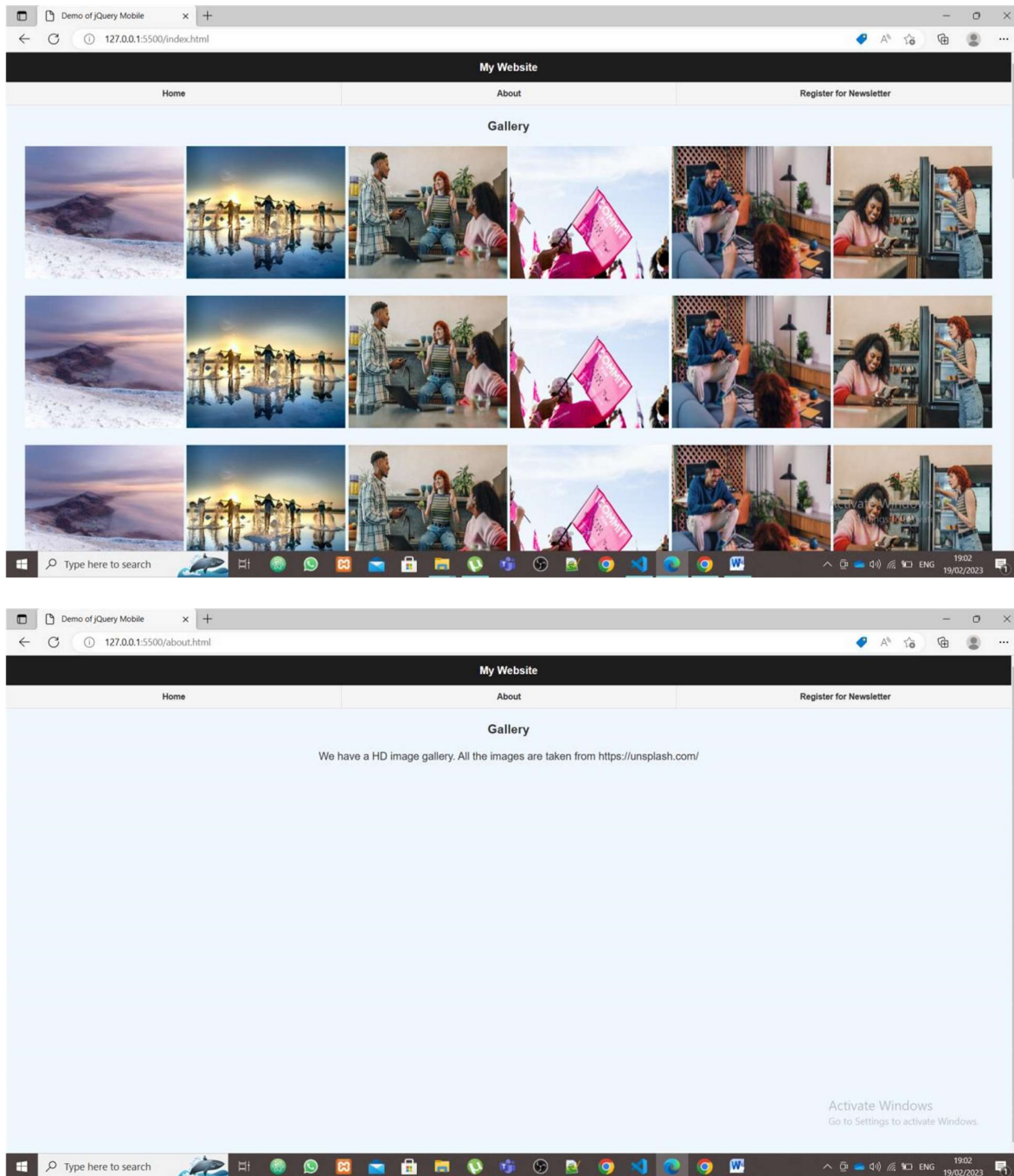
    </center>

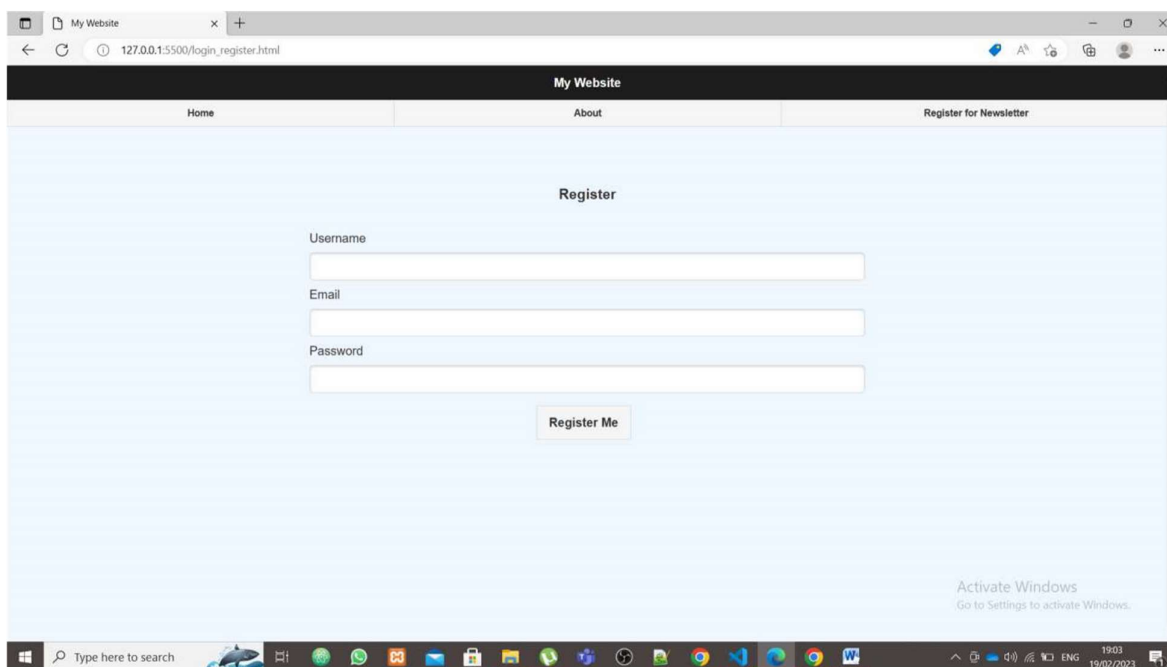
  </div>

</div>

</body>

</html>
```





Conclusion: Thus, we are able to create a simple Mobile Website using jQuery Mobile.